# Module 4: An Entry-Level Analyst's Toolkit & Playbooks

## An Entry-Level Analyst's Toolkit

Every organization provides a different toolkit depending on its specific security needs [cite: 64]. As a future analyst, it is essential to be familiar with industry-standard tools and demonstrate the ability to learn similar tools in any workplace [cite: 64].

## Understanding Logs

- **Definition:** Logs are records of events that occur within an organization's systems, such as employee sign-ins, service requests, or web service access [cite: 63, 64].
- **Purpose:** They help security professionals identify vulnerabilities, detect anomalous activity, and uncover potential security breaches [cite: 63].

## Security Information and Event Management (SIEM) Tools

A **SIEM tool** is an application that collects, aggregates, and analyzes log data to monitor critical activities and detect security incidents in real time [cite: 63, 64].

- **Automated Triage:** Filtering through massive volumes of log data manually could take hours or days; SIEM tools drastically reduce data overload by generating automated alerts for specific risks, threats, and vulnerabilities [cite: 64].
- **Dashboards:** SIEM tools provide visual dashboards that categorize data, allowing analysts to customize views and prioritize investigations [cite: 64].
- **Deployment & Hosting:** SIEM solutions come with various hosting options, including on-premises and cloud-hosted architectures [cite: 64]. Cloud-hosted SIEMs tend to be easier to deploy, scale, and maintain, making them attractive for smaller or less experienced teams [cite: 64].
- **Industry Examples:** Leading platforms include *Splunk Enterprise* and Google's *Chronicle*, both used to filter data across hybrid environments to proactively mitigate threats [cite: 63].

## Network Protocol Analyzers (Packet Sniffers)

A **network protocol analyzer** (also called a **packet sniffer**) captures and inspects data packets traversing a network in real time [cite: 63, 64].

- **Functionality:** Keeps a comprehensive record of network communications encountered by connected devices, assisting analysts in identifying unauthorized connections or data exfiltration [cite: 64].
- **Common Tools:** Prominent examples include command-line utilities like `tcpdump` and graphical analyzers like *Wireshark* [cite: 63].

## Playbooks in Security Operations

A **playbook** is a documented manual that provides step-by-step instructions for operational procedures, guiding analysts on how to respond consistently to security incidents [cite: 63, 65].

*Practical Scenario:* An analyst investigating a breach at a medical clinic must collect forensic findings to present to a cyber insurance carrier. The insurance payout relies heavily on adhering strictly to standard forensic playbooks [cite: 65]:

- **Chain of Custody Playbook:**
  - **Definition:** The formal process of documenting evidence possession, custody, and transfer throughout the incident lifecycle [cite: 65].
  - **Execution:** Forensic analysts must record the *who, what, where, when,* and *why* for all collected digital evidence [cite: 65].
  - **Accountability:** Evidence must remain tamper-proof, tracked, and securely stored; every physical or logical transfer must be logged so investigators can verify its integrity in court or during audits [cite: 65].
- **Protecting and Preserving Evidence Playbook:**
  - **Fragile & Volatile Data:** Digital evidence can be easily modified or lost permanently if a device shuts down or reboots [cite: 65].
  - **Order of Volatility:** Analysts must follow the order of volatility—a preservation sequence prioritizing the most ephemeral data first (e.g., CPU cache, RAM/routing tables) down to non-volatile secondary storage (hard drives, archival backups) [cite: 65].
  - **Forensic Best Practices:** Because improper handling compromises digital evidence and renders it inadmissible, analysts preserve master images and perform analyses solely on verified forensic copies [cite: 65].